

SECTION A

Answer all questions in the spaces provided.

1. Give the electronic configuration of a copper atom. [1]

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2. Excess sodium hydroxide is added to an aqueous solution of chromium(III) chloride, CrCl_3 . Give the formula of the chromium-containing ion formed. [1]

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3. (a) Write the equation for the reaction of hot concentrated aqueous sodium hydroxide with chlorine. [1]

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- (b) This is an example of a disproportionation reaction. State what is meant by the term *disproportionation*. [1]

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4. Give the observation(s) expected when water is added to SiCl_4 . [1]

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5. A 25.0 cm³ sample of a solution containing iodine is titrated against a sodium thiosulfate solution of concentration 0.0200 mol dm⁻³. This requires 23.25 cm³ of sodium thiosulfate for complete reaction.



Calculate the concentration of the iodine in the solution in g dm⁻³.

[2]

Concentration = g dm⁻³

6. The enthalpy of solution of sodium chloride is +4 kJ mol⁻¹. Explain why this compound is soluble in water despite this value being positive. [1]

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7. When a solution containing the weak base ammonia is neutralised using the strong acid sulfuric acid, a solution of ammonium sulfate is formed. Suggest a pH for this solution, giving a reason for your answer. [2]

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